

Question 1: Create a Dockerfile for a FastAPI Application

Objective: Write a simple Dockerfile to containerize a FastAPI application.

Task:

1. Create a directory named `fastapi-app` on your host machine.
2. Inside the directory, create a simple FastAPI application:
 - Create `main.py` with a basic API endpoint.
 - Create `requirements.txt` with FastAPI and Uvicorn.
3. Write a **Dockerfile** that:
 - Uses `python:3.9-slim` as the base image.
 - Sets the working directory to `/app`.
 - Copies `requirements.txt` and installs dependencies.
 - Copies the application code.
 - Exposes port `8000`.
 - Runs the FastAPI app using Uvicorn.
4. **Build** the image with the tag `fastapi-app:1.0`.
5. **Run** a container from the image, mapping port `8000` on your host to port `8000` in the container.
6. Access the API at `http://localhost:8000` and verify it works.

Sample Files for Students:

main.py:

```
from fastapi import FastAPI

app = FastAPI()

@app.get("/")
def read_root():
    return {"message": "Hello from Docker!"}

@app.get("/hello/{name}")
def say_hello(name: str):
    return {"message": f"Hello {name}!"}
```

requirements.txt:

Docker Training

fastapi
uvicorn

Hints:

Step 1 - Create directory and files:

```
mkdir fastapi-app  
cd fastapi-app  
# Create main.py and requirements.txt (use the sample above)
```

Step 2 - Write Dockerfile:

```
FROM python:3.9-slim  
WORKDIR /app  
COPY requirements.txt .  
RUN pip install -r requirements.txt  
COPY main.py .  
EXPOSE 8000  
CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
```

Step 3 - Build the image:

```
docker build -t fastapi-app:1.0 .
```

Step 4 - Run the container:

```
docker run -d -p 8000:8000 --name fastapi-container fastapi-app:1.0
```

Step 5 - Test the API:

```
curl http://localhost:8000  
# Output: {"message":"Hello from Docker!"}
```

```
curl http://localhost:8000/hello/Student  
# Output: {"message":"Hello Student!"}
```